

How a band feature is actually computed

From a sampled wave to e_{1x} , e_{2x} , e_{bpfo} and e_{hi} — by FFT, or by Goertzel

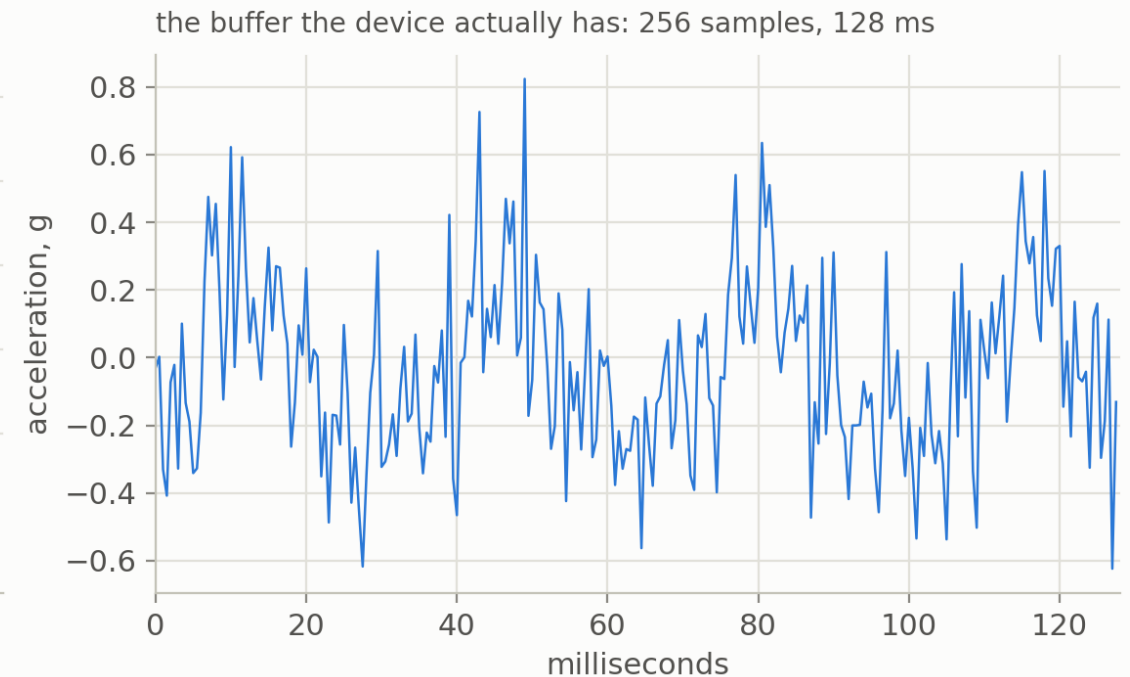
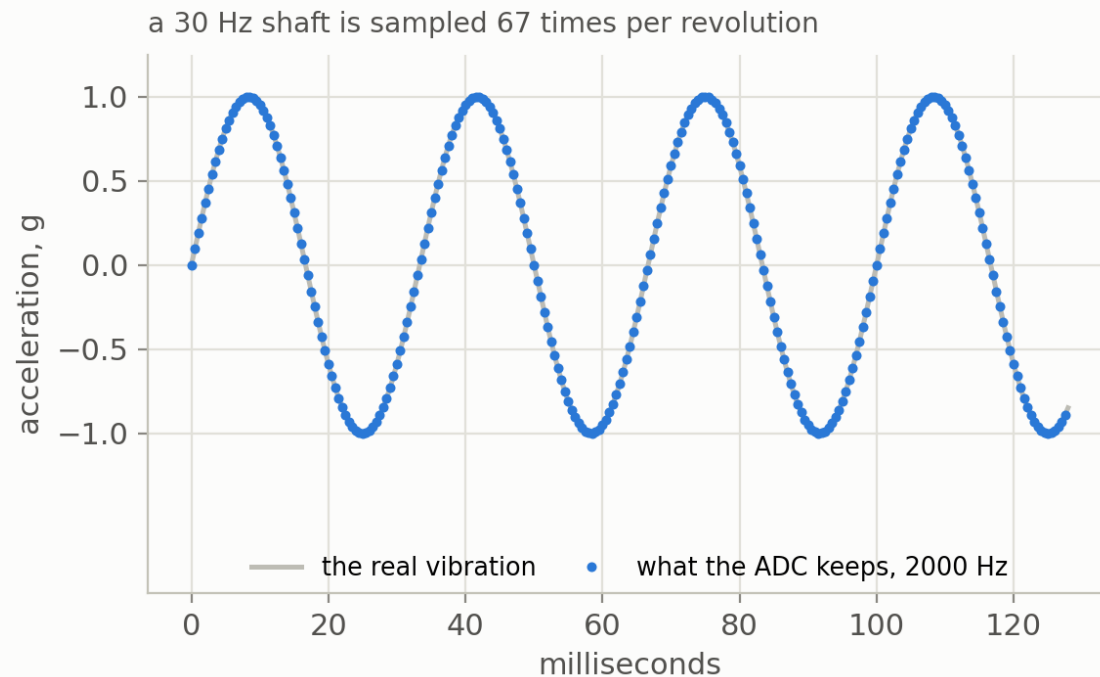
- What a sample rate buys you, and the one mistake no software can undo
- The DFT as a bank of correlations — two probes per bin, and why
- Bin width is $f_s/N = 7.8125$ Hz, and it is not a preference
- Spectral leakage, and the window function that matters more than the algorithm
- The FFT: 132,096 multiplies down to 8,450
- Goertzel: one bin, 260 multiplies, two floats of state
- Choosing between them by counting bins — and why e_{hi} settles it

Notebook: `Lecture8_Supplement1_Frequency_Features.ipynb`

Supplementary material — background for Lecture 8, not new examinable content

A sample is a number; a rate is a promise

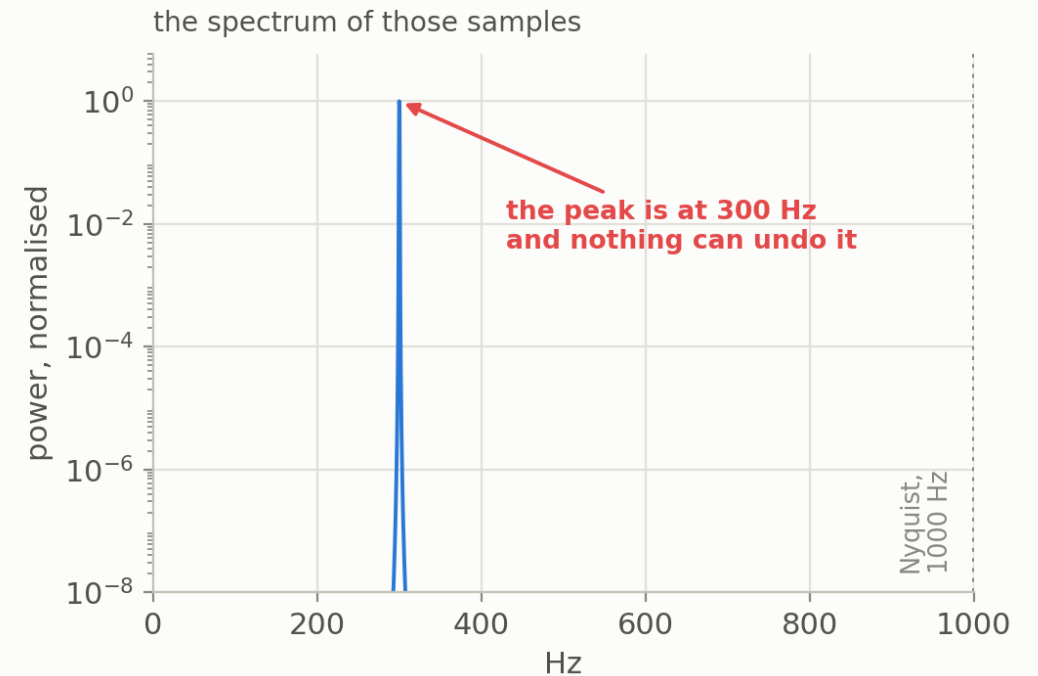
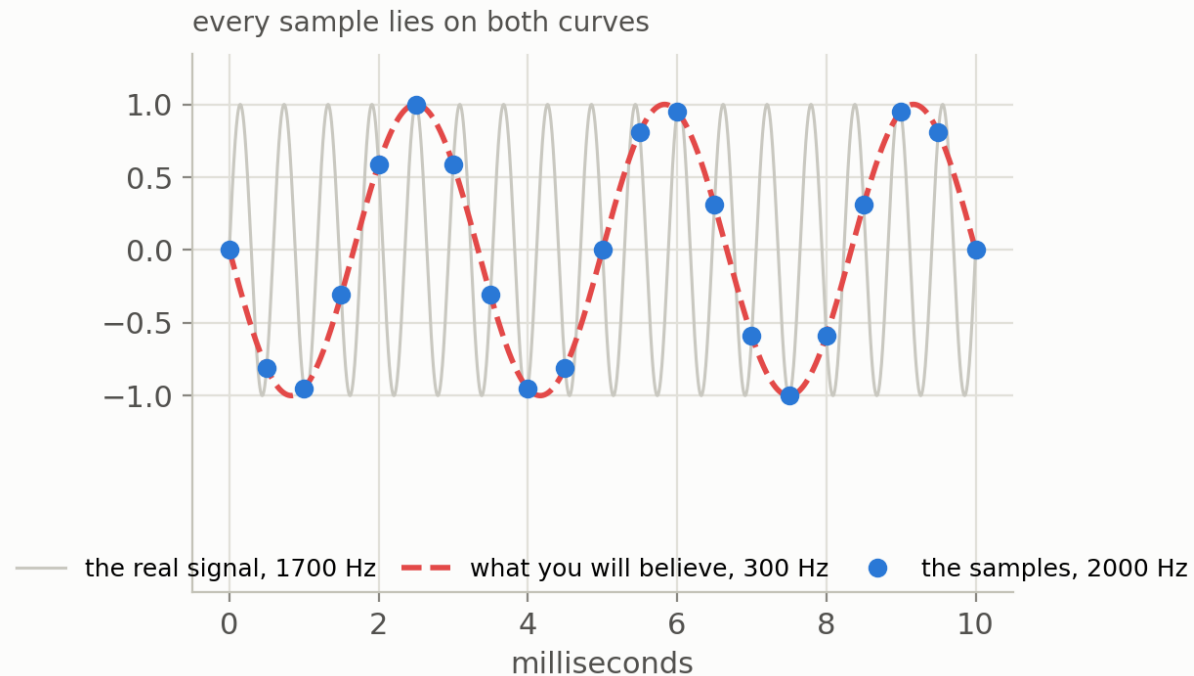
- The ADC reads the accelerometer 2000 times a second and throws everything between readings away.
- At 30 Hz shaft speed that is 67 samples per revolution — but the rate was chosen for the 600 Hz housing resonance, not the shaft.



256 samples at 2000 Hz is 128 ms of machine — one window, 512 bytes, and the unit every feature in this course is computed on.

The one mistake no software can undo

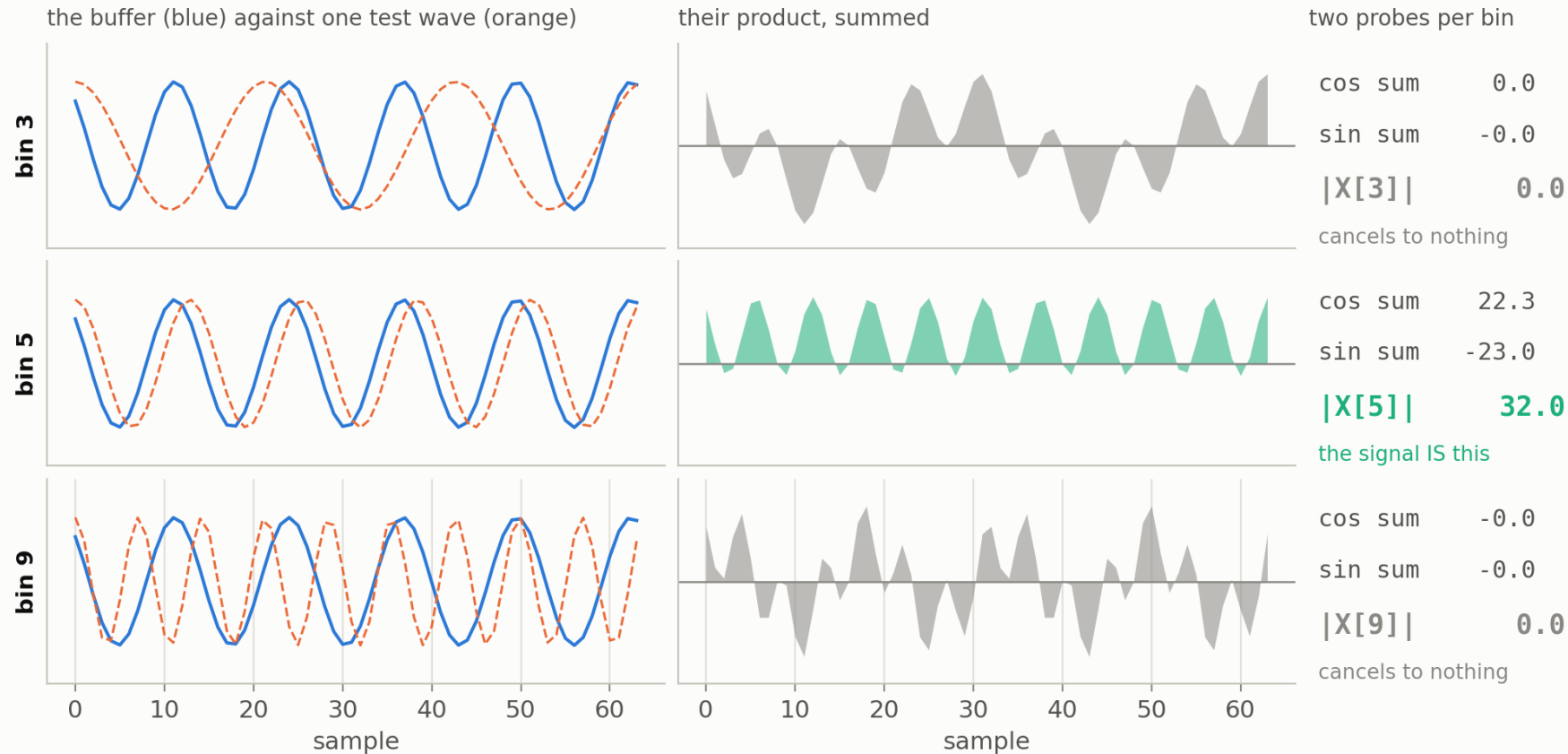
- A 1700 Hz vibration sampled at 2000 Hz does not disappear — it moves to 300 Hz and sits there looking like a real feature.



The anti-alias filter goes BEFORE the ADC. After it, the information you needed is already gone.

The DFT is a bank of correlations

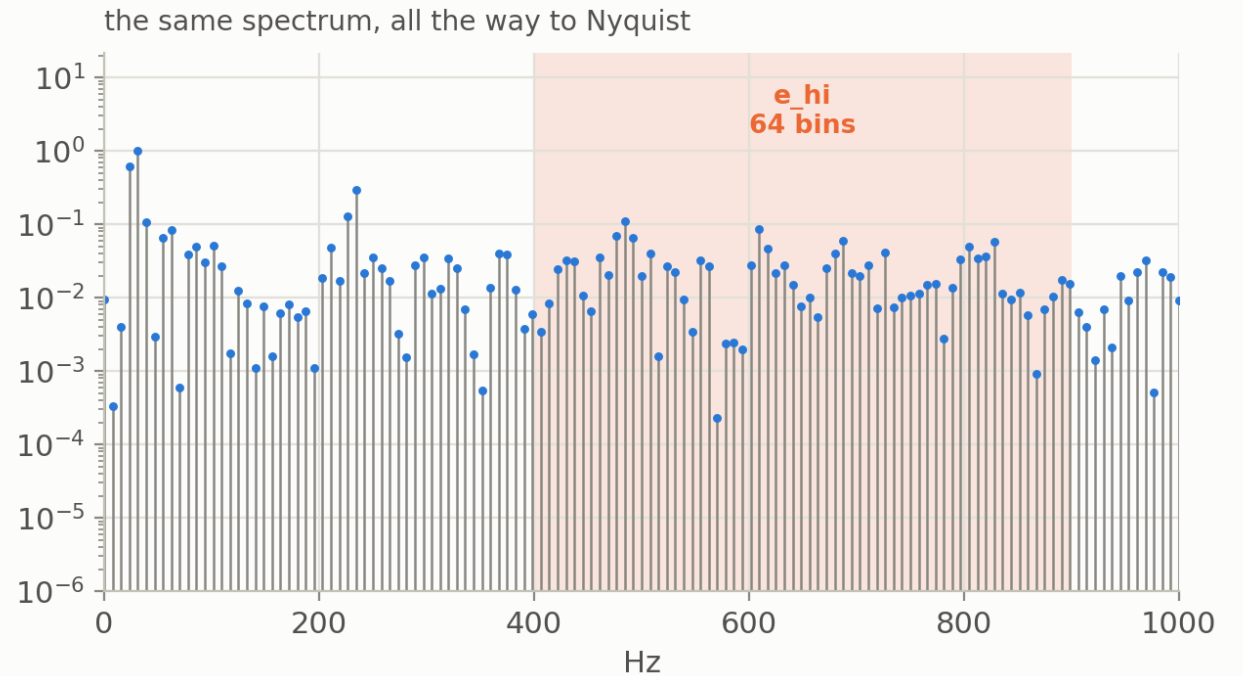
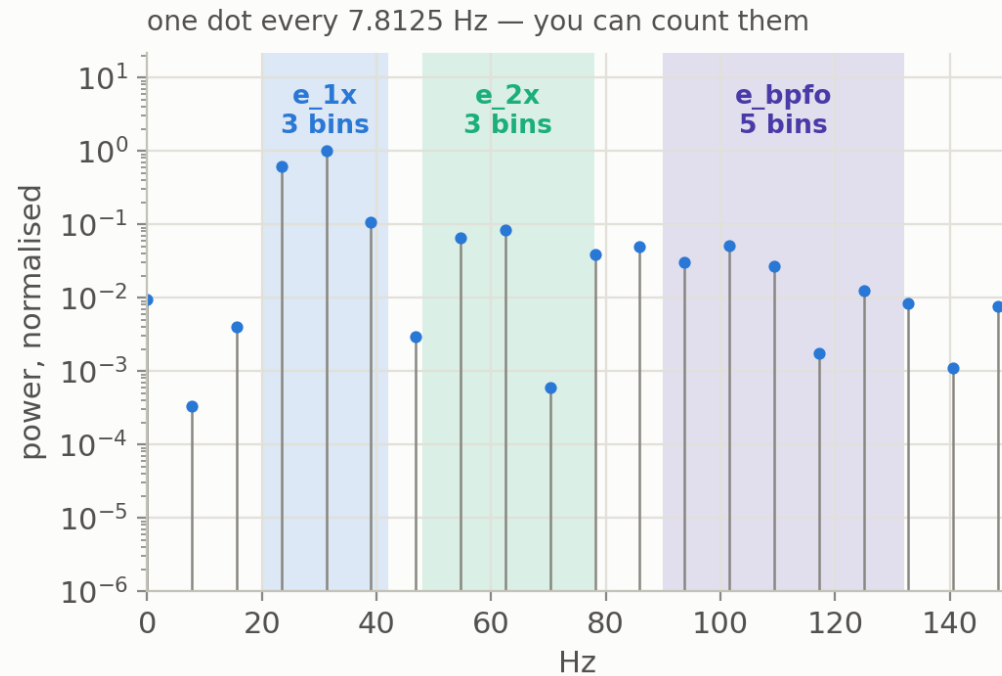
- Multiply the window by a test wave, sample by sample, and add up the result. If the wave is in your signal the products line up; if not, they cancel.



Two probes per bin — cosine and sine — because one probe would report zero for a signal a quarter-cycle out of step. The magnitude is what survives.

The bins are not yours to choose

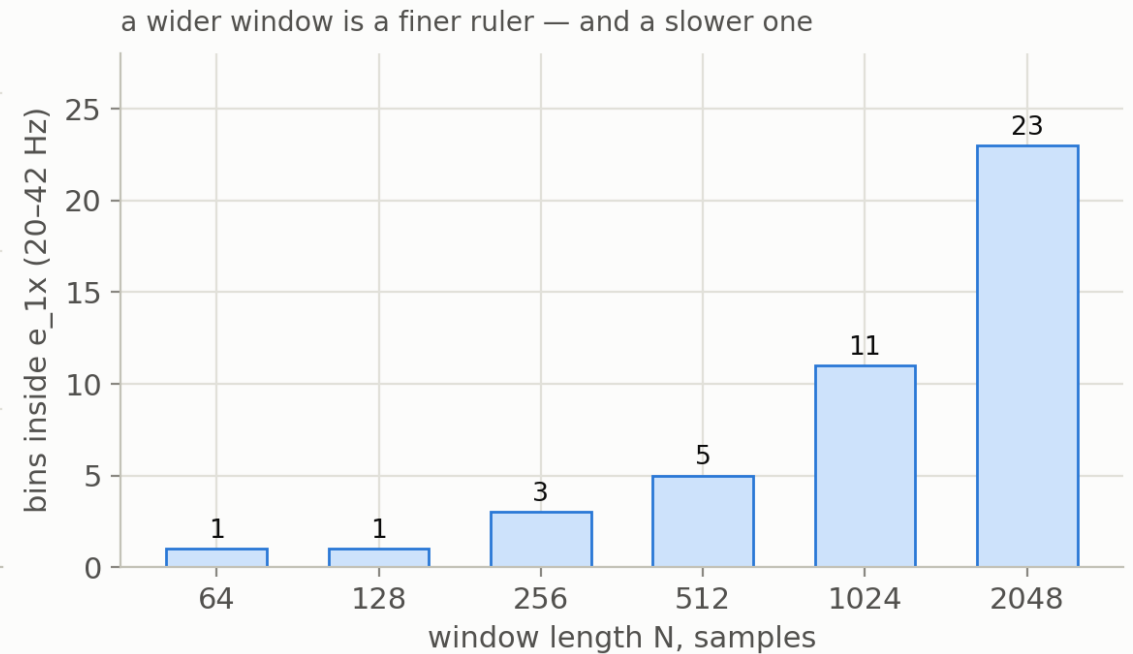
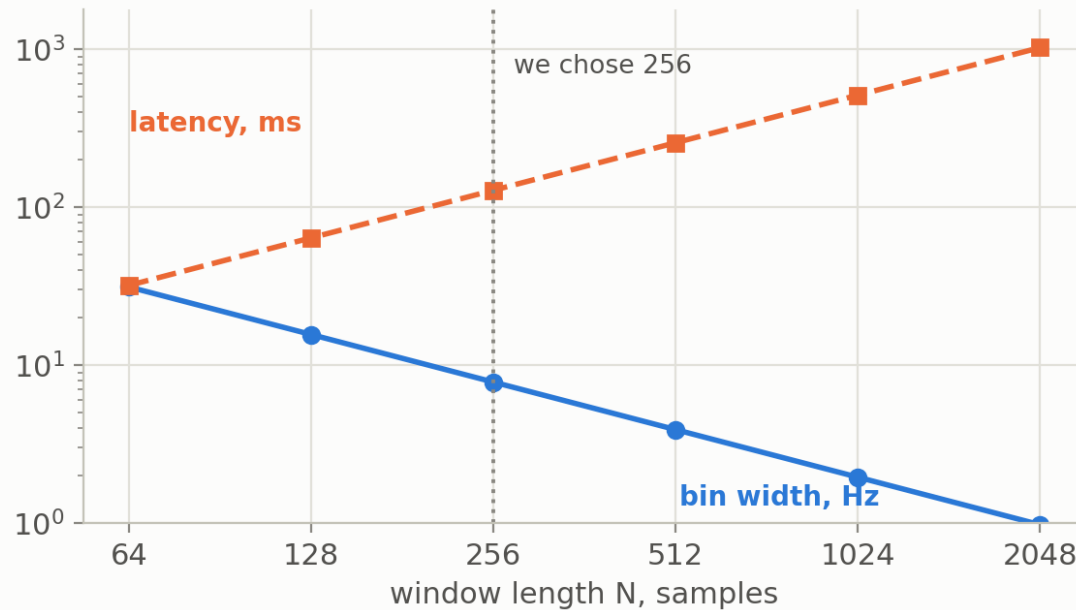
- A test wave must fit a whole number of cycles in the window, so bin k sits at $k \cdot f_s / N$. Nothing exists between neighbouring bins.
- $\Delta f = 2000 / 256 = 7.8125$ Hz — you cannot ask this window what is happening at 30.0 Hz.



This is why a band and not a line: at 7.81 Hz resolution, with the shaft wandering 28–32 Hz and no tachometer, one bin would miss it half the time.

A finer ruler is a slower one

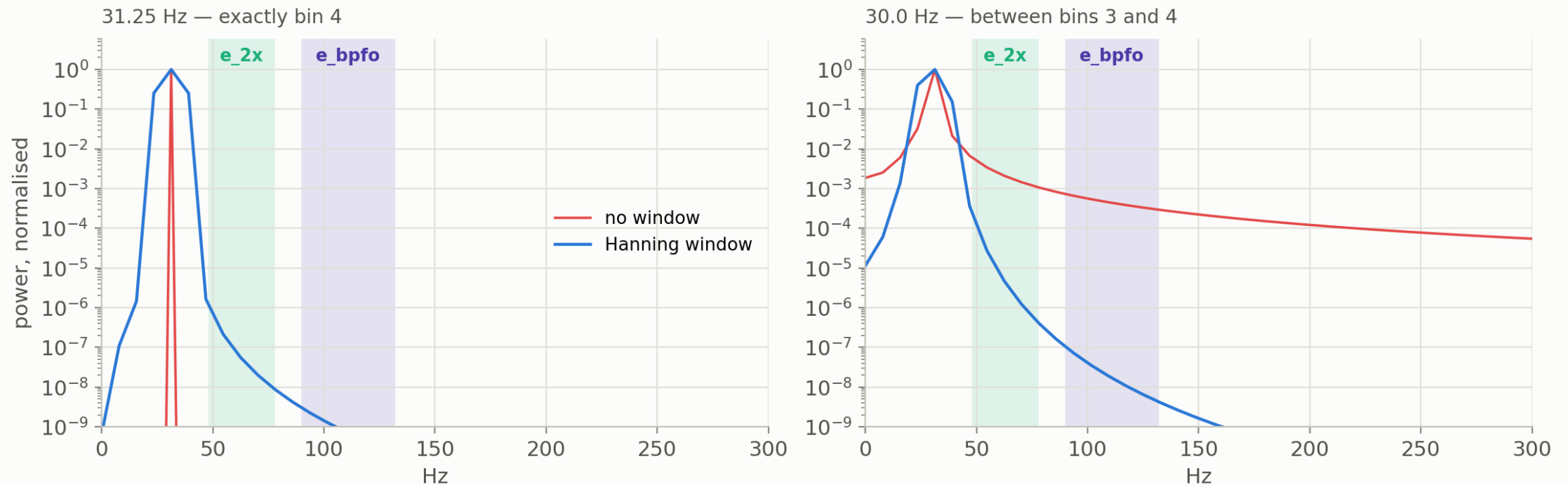
- Bin width and latency are the same number read two ways: double N , halve Δf , double the delay before the device can say anything.



Lecture 8's sweep showed accuracy still rising at $N = 1024$. We chose 256 anyway — because 128 ms is the latency the actuator can live with, and 512 bytes is the RAM we can spare.

Leakage, and the line that fixes it

- A 30.0 Hz shaft is 3.84 bins — between bin 3 and bin 4. It does not fit a whole number of cycles, so the window wraps with a step, and a step contains every frequency.



A pure shaft tone, on a perfectly healthy machine, puts 0.22 % of its energy into e_bpfo with no window — 23,627× more than with one.

The FFT: the same answer, far fewer operations

- The definition needs N multiply-accumulates for each of $N/2+1$ bins — 132,096 real multiplies per window, which interpreted MicroPython will not do 15 times a second.

```
# radix-2, decimation in time – the routine Lecture 11 runs on the ESP32
bit_reverse(re, im)           # pure addressing, no arithmetic
for size in (2, 4, 8, ..., N): # log2(N) = 8 passes
    for each butterfly in the pass:
        t = twiddle * x[b]
        x[b] = x[a] - t
        x[a] = x[a] + t
```

What it costs

8,450 real multiplies and 512 floats of scratch space, every window — whether you wanted one bin or all 129.

What it gives

Every bin, exactly. Our version agrees with `numpy.fft` to $1e-13$, and it is the same transform — the FFT is a shortcut, not an approximation.

132,096 → 8,450 multiplies, and not one digit of the answer changes.

Goertzel: one bin, two variables

- A second-order recurrence — literally an IIR filter tuned to one bin's frequency. Run it over the window and the last two states hold the answer.

```
coeff = 2 * cos(2 * pi * k / N)
s1 = s2 = 0.0
for v in buf:                                # ONE real multiply per sample
    s0 = v + coeff * s1 - s2
    s2, s1 = s1, s0

power = s1*s1 + s2*s2 - coeff*s1*s2
```

What it costs

260 real multiplies per bin, and two floats of state. No complex arithmetic, no twiddle table, no output array.

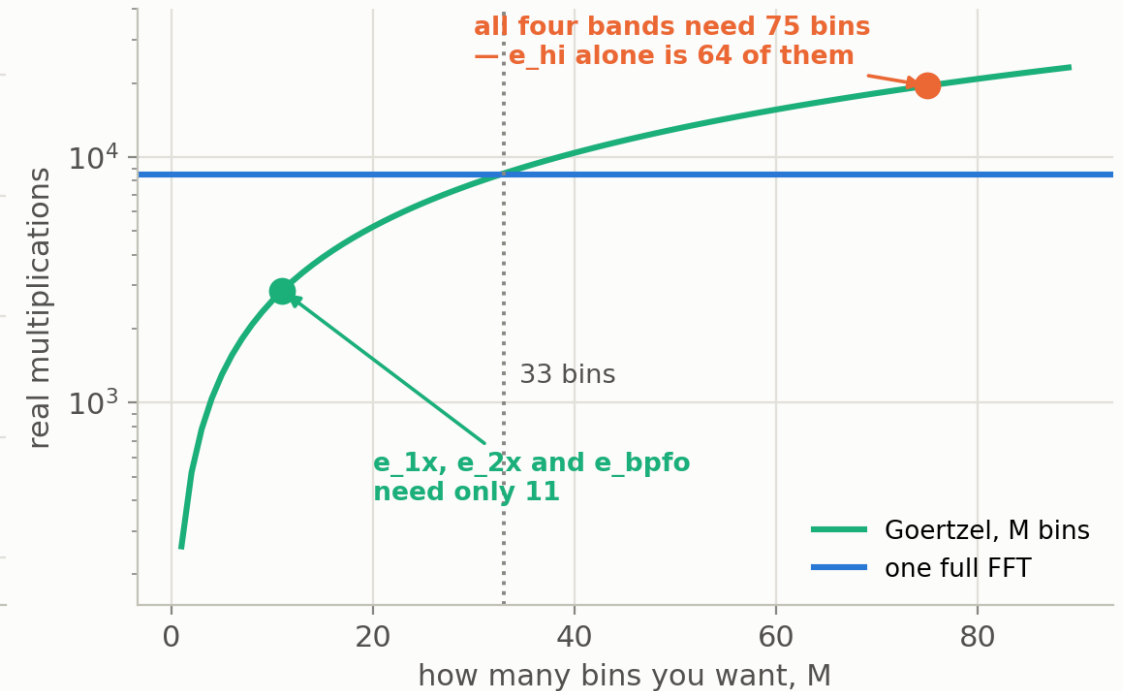
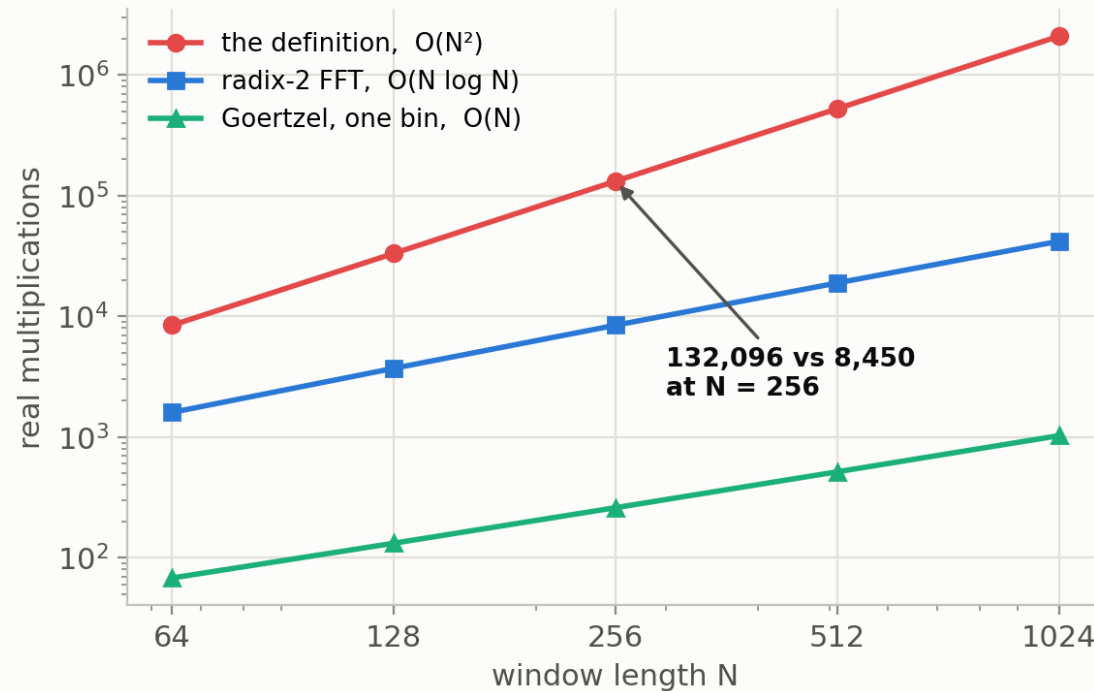
What it gives

One bin. Agrees with the FFT to 1e-14 on the same window — same transform, different route.

Goertzel has no opinion about windows: window the buffer before the loop, exactly as for the FFT, and it matches the notebook.

So which one? Count the bins.

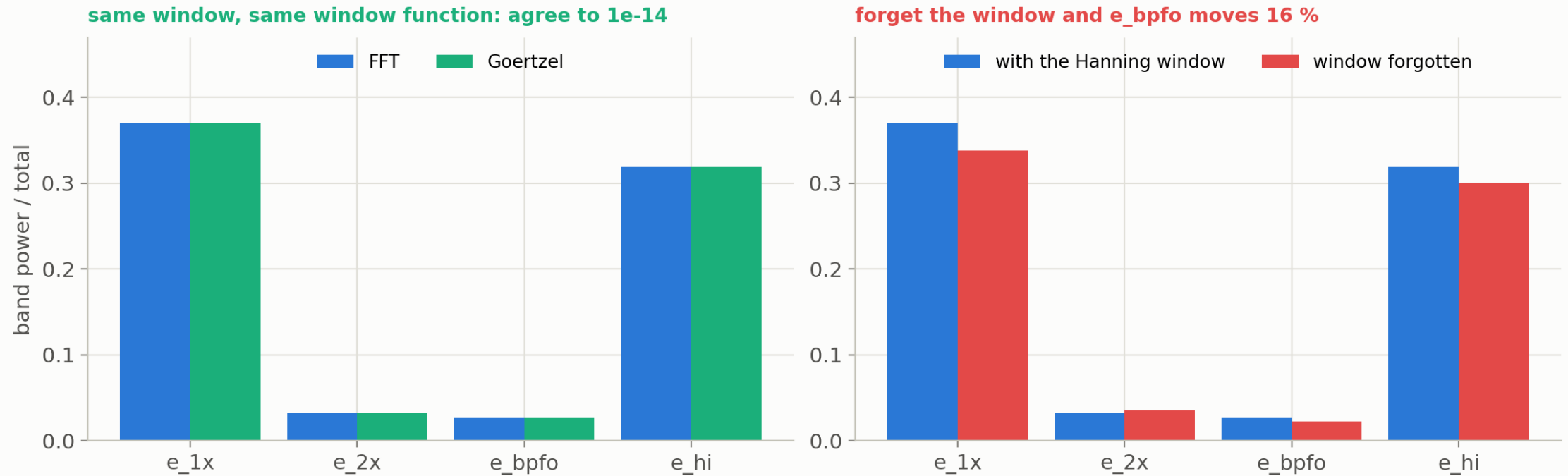
- One FFT is 8,450 multiplies. Goertzel is 260 per bin. Up to 32 bins Goertzel is cheaper; from 33 the FFT is, and every bin is then free.



Our four bands need 75 bins — e_hi alone is 64 of them. Drop e_hi and the other three need 11, where Goertzel wins by 3.0x.

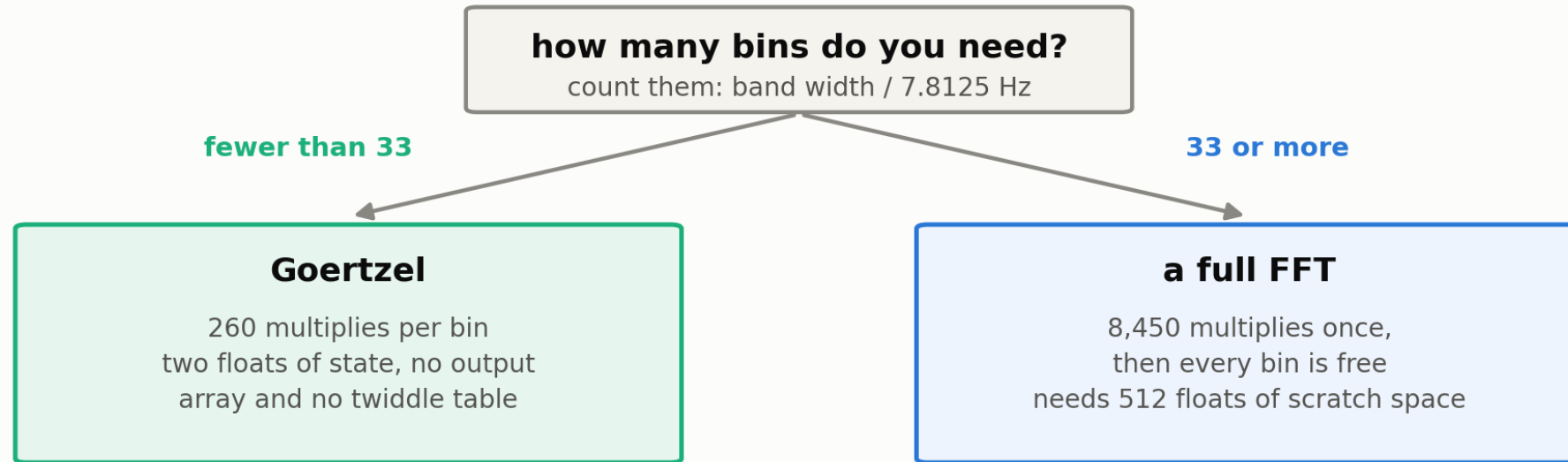
They agree; the window is what changes the answer

- Same window, same window function, two completely different routes — and twelve identical numbers.



Forget the window and e_bpfo moves by 16 % — larger than the healthy-to-faulty change in crest that Lecture 8 relies on.

What this means for your device code



our four bands need 75 bins, because e_hi alone is 64 of them

drop e_hi and the other three need only 11 — then Goertzel wins by 3.0x

If you want e_1x, e_2x, e_bpfo

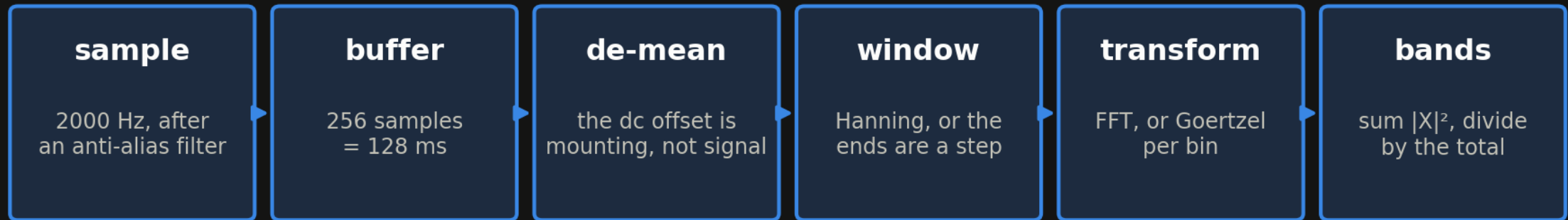
Goertzel, 11 bins, 2,860 multiplies, two floats of state.

If you want e_hi or dom_freq

One FFT: 8,450 multiplies, 512 floats of scratch. Lecture 11 measures 923 μ s.

De-mean, then window, then transform. Two lines, and they matter more than the algorithm underneath them.

Six steps, and the two that get left out



the two middle steps are the ones students leave out, and they change the answer more than the algorithm does

1. The sample rate is a promise you cannot renegotiate — filter before the ADC, not after.
2. Bin width is $f_s/N = 7.8125$ Hz. It is arithmetic, not a setting.
3. A band, not a line, whenever the shaft can wander and you have no tachometer.
4. De-mean, then window, then transform. These two lines outweigh the algorithm.
5. FFT and Goertzel give the same numbers (1e-14). If yours differ, you changed the window.
6. Count bins to choose: under 33, Goertzel; over, one FFT. `e_hi` alone is 64.